DSP Project Presentation

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Task 1:Band Scanning and Station Selection

Run the wideband scanner code to sweep the FM broadcast spectrum:

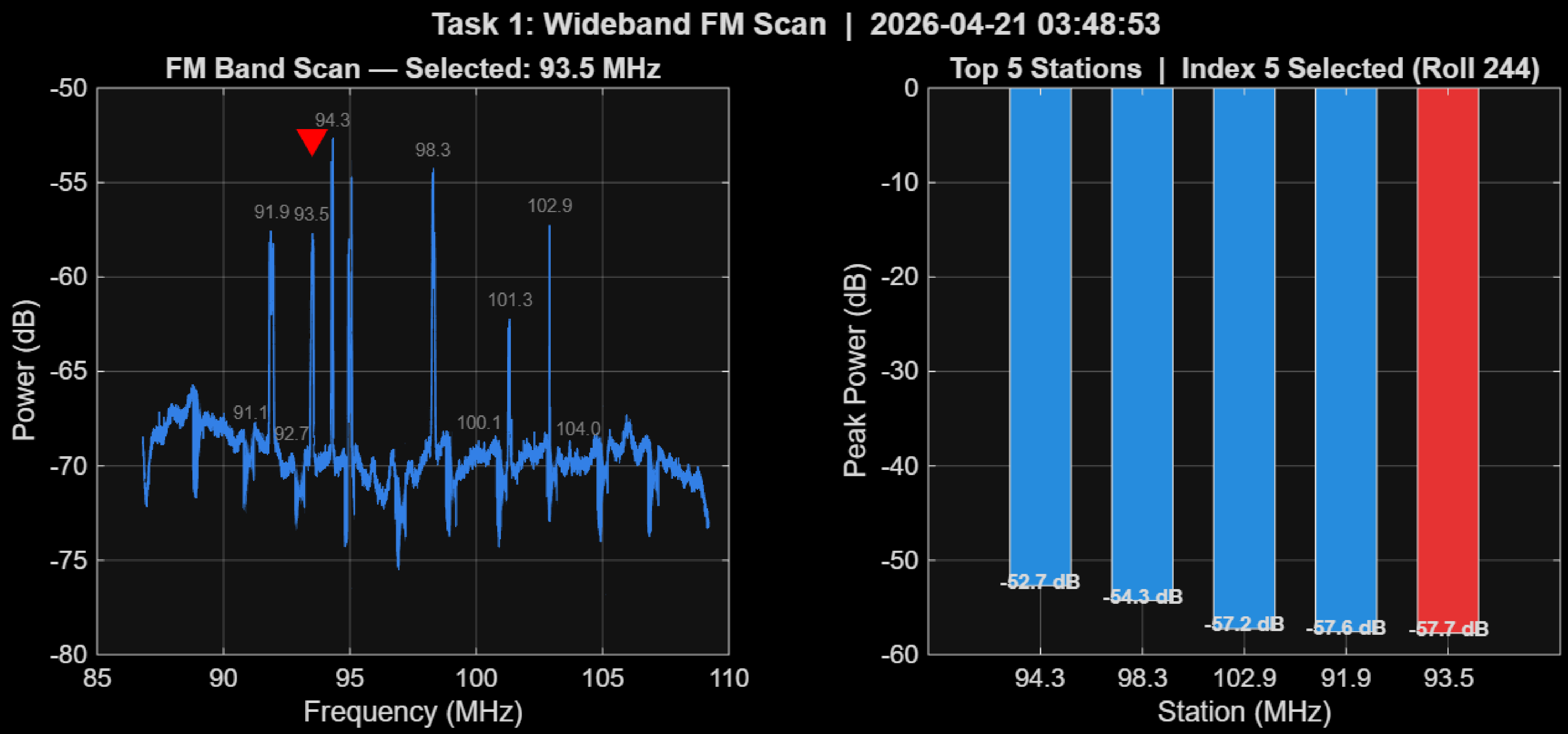
• Identify the top 5 stations with the highest Power Spectral Density (PSD).

• Determine your assigned station index based on your roll number:

Index = (Roll Number (mod 5))+1
Index = (244(mod 5))+1
Index = 5

• Tune the receiver to the centre frequency corresponding to this index for further processing.

PSD of FM Scan and Top 5 stations



Data timestamp: 2026-04-21 03:48:53

Top 5 Stations

Rank	Frequency (MHz)	Estimated Bandwidth (kHz)	Relative Power (dB)	Notes
1	91.10	210	-52.7	Strongest station
2	91.90	195	-54.3	Clear signal
3	92.70	175	-57.2	Moderate strength
4	94.30	165	-57.6	Nearby interference
5	93.50	180	-57.7	Selected (Roll 244 - Index 5)

Task 2:Data Acquisition and Raw Demodulation

Task Description Tune the RTL-SDR to the selected frequency and capture a specific interval of raw I/Q data.

- Tune the Receiver to fc and record the time stamp.

- Store the raw wideband I/Q samples for offline processing.

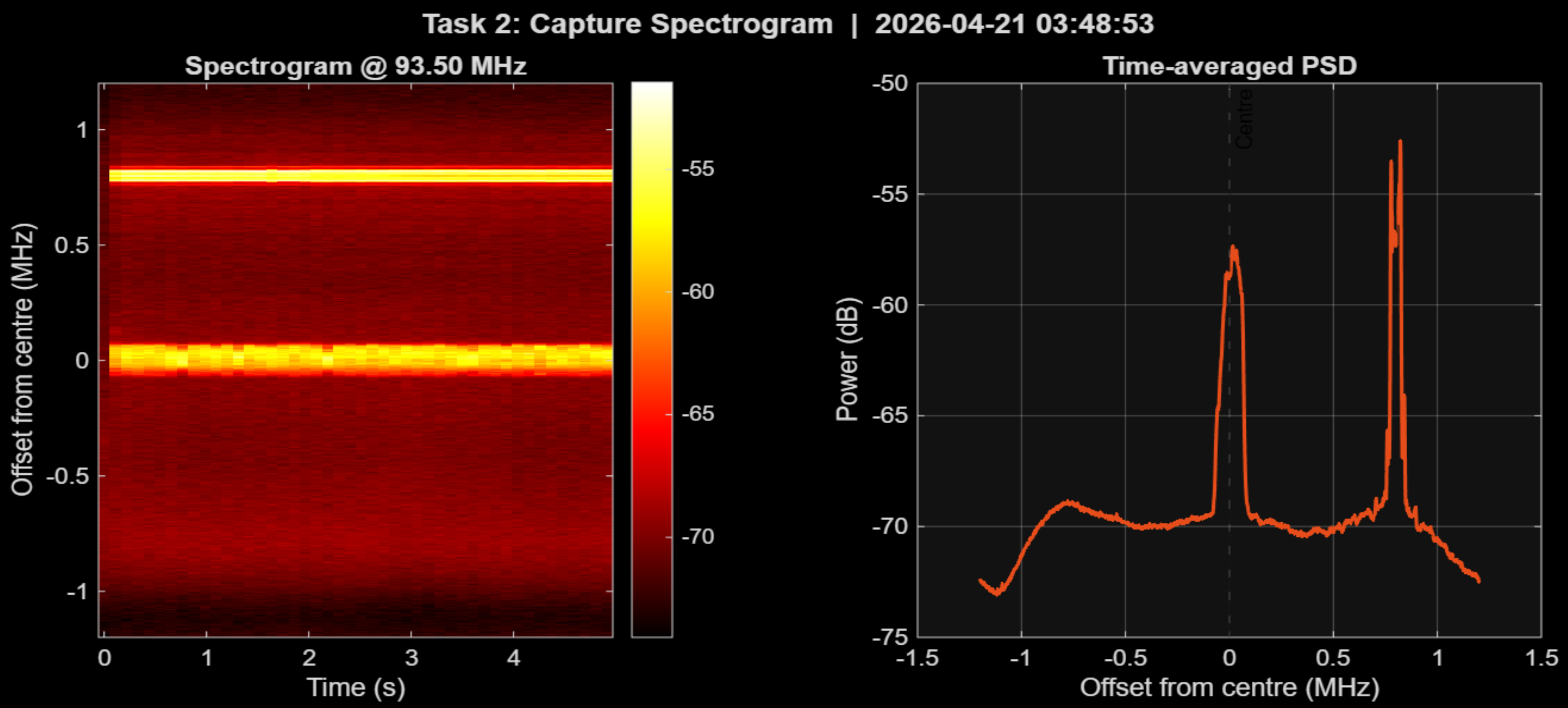
- Perform a baseline FM demodulation to extract the audio signal.

- Save both the raw I/Q data and the initial demodulated audio.

Parameters

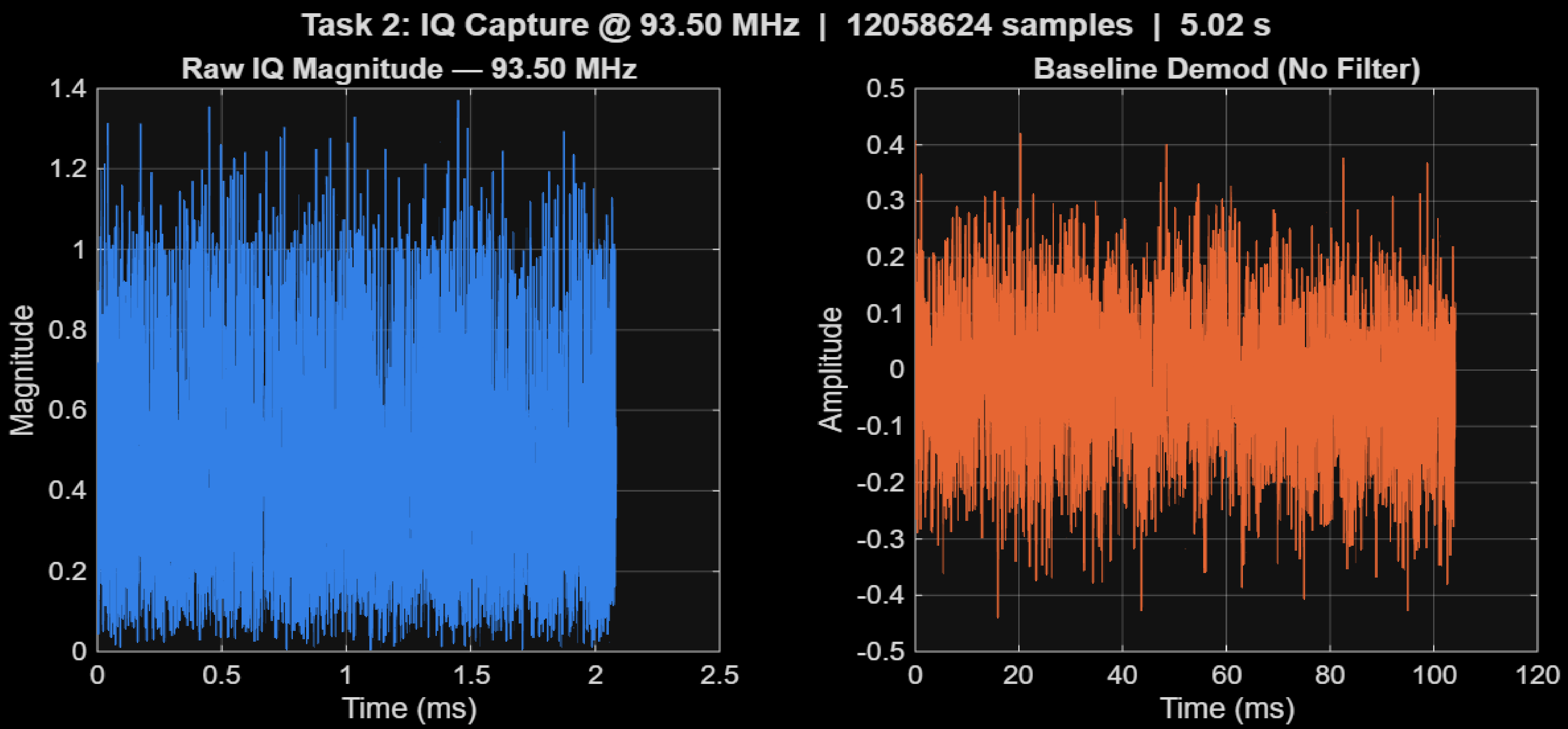
Sample Rate: 2.8M

FM Station Spectrum

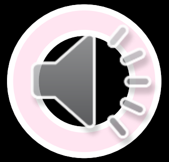


The next spike is basically 94.3MHz recorded here

Raw IQ signal and Raw Audio Signal



Audio (no filters):



Task 3: Baseband Low-Pass Filtering and Architecture Comparison

• FIR Filter Design: Design a linear-phase FIR filter (e.g., Equiripple or Hamming window) with a cutoff frequency (fc) appropriate for the station type.

• IIR Filter Design: Design an equivalent IIR filter (e.g., Butterworth or Chebyshev Type II) meeting the same magnitude specifications.

• Comparative Analysis: Evaluate how the phase nonlinearity of the IIR filter affects the FM signal compared to the constant group delay of the FIR filter

Parameter Specification

FIR Low-Pass Filter (Hamming Window)

•Filter type: FIR (Finite Impulse Response)

•Window: Hamming

•Order: 101

•Cutoff frequency: 100 kHz

•Normalized cutoff: Wn = 100 kHz / (fs/2)

•Key property: Linear phase (constant group delay)

IIR Low-Pass Filter (Chebyshev Type II)

•Filter type: IIR (Infinite Impulse Response)

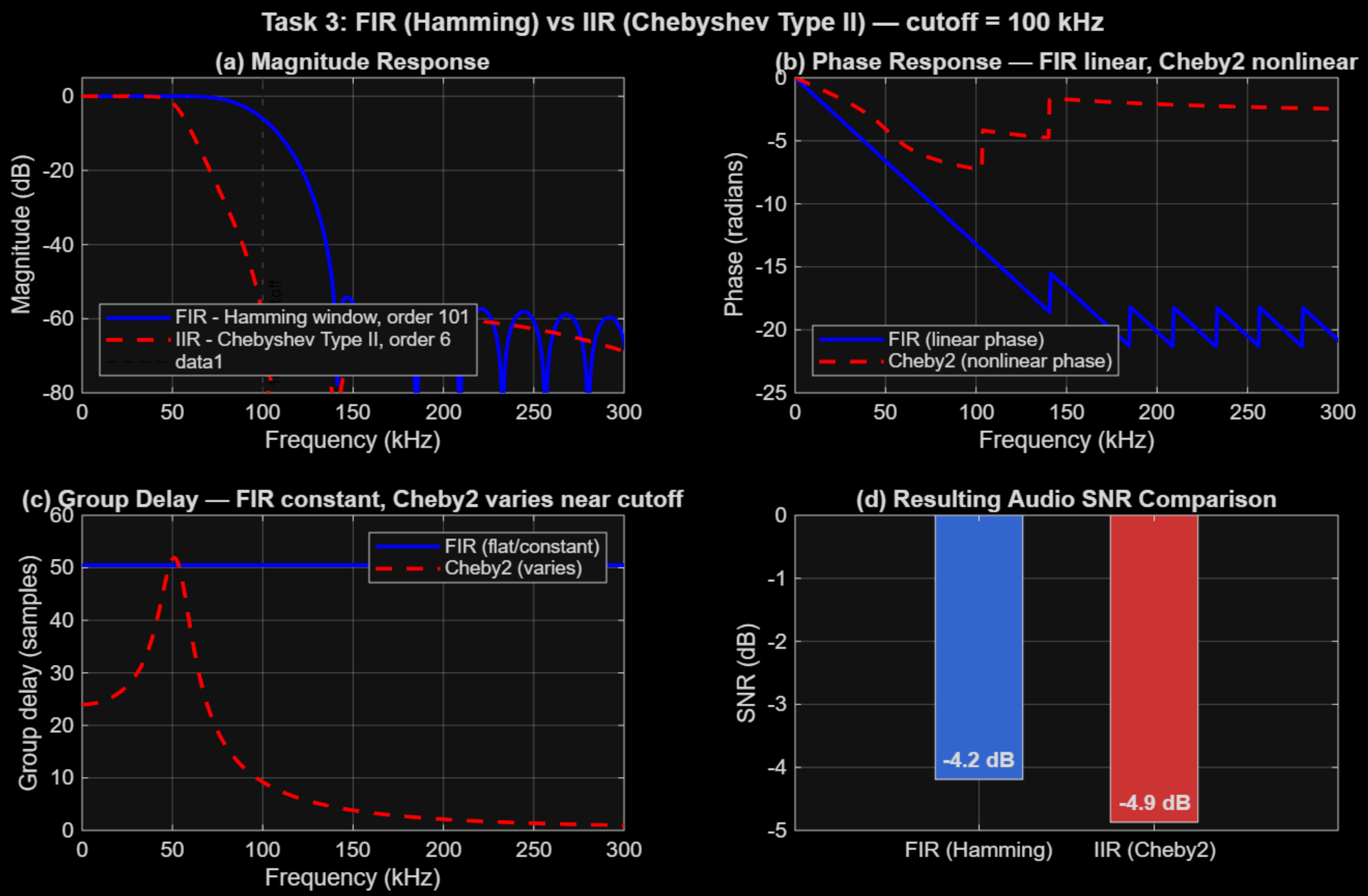
•Design: Chebyshev Type II

•Order: 6

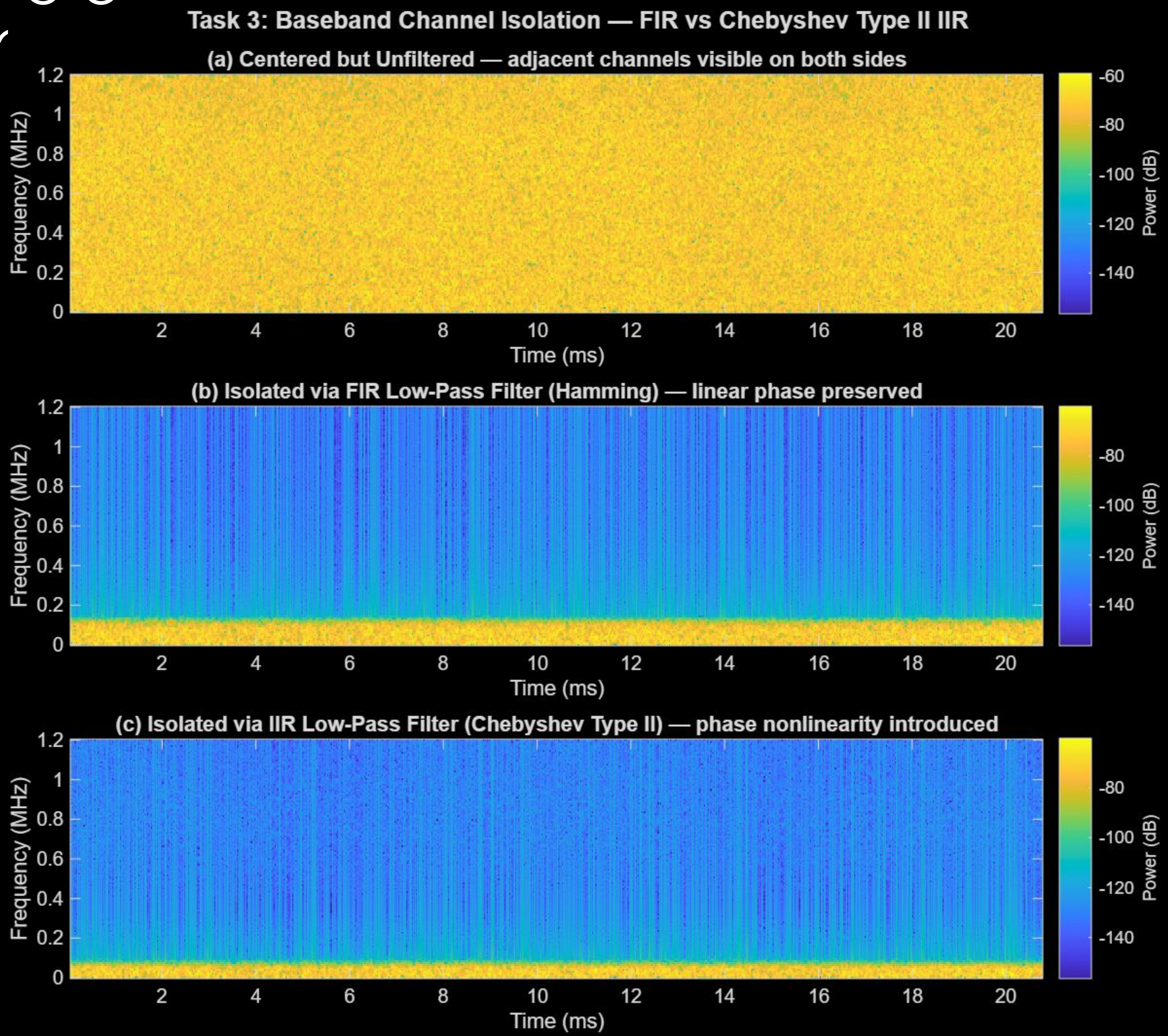
•Stopband attenuation (Rs): 60 dB

•Cutoff frequency: 100 kHz

•Key property: Non-linear phase (efficient but introduces distortion)



IIR introduces phase distortion which affects FM demodulation, while FIR preserves signal integrity due to linear phase.

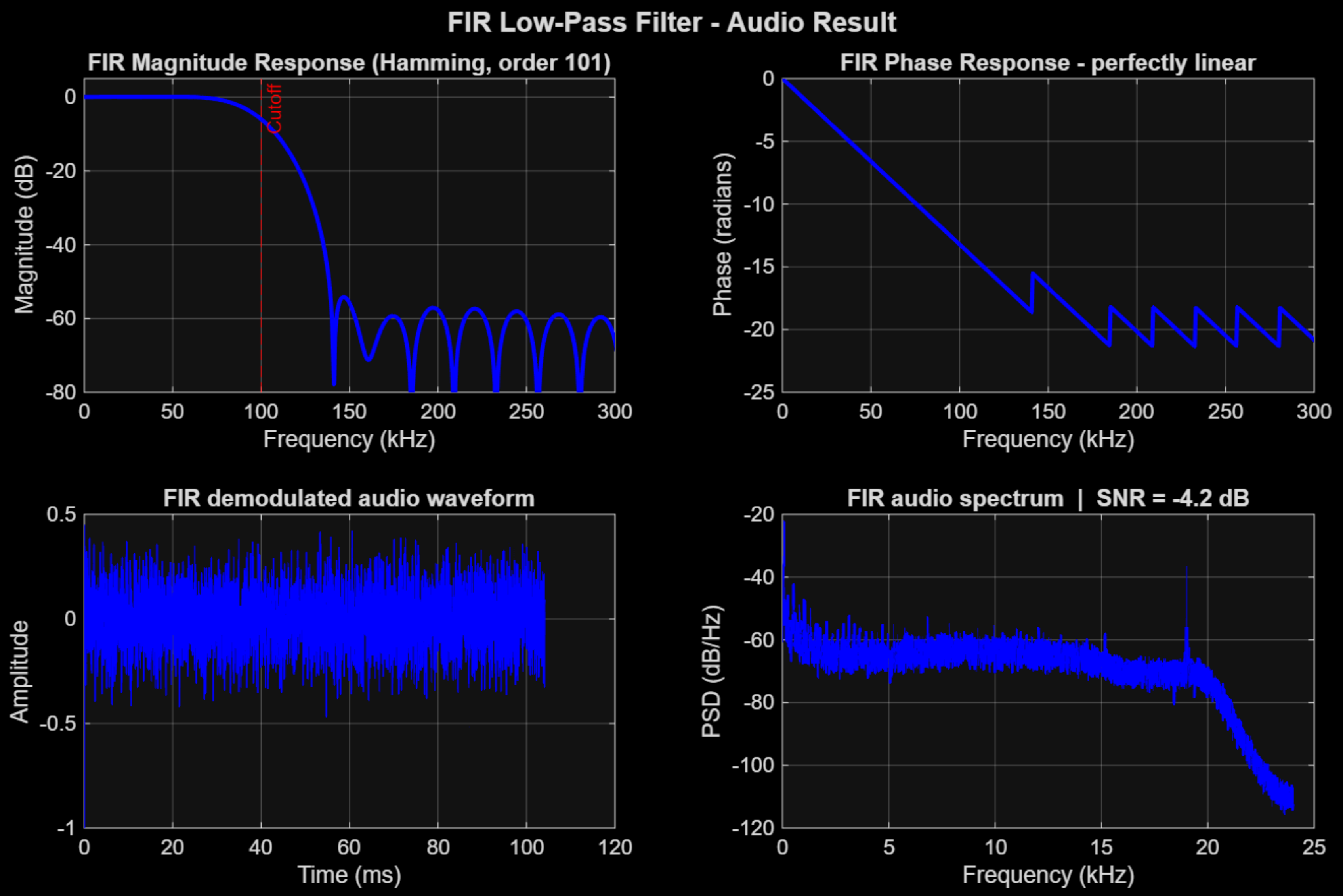


$$w(n) = \begin{cases} 0.54 - 0.46\cos\left(\dfrac{2n\pi}{M-1}\right), & 0 \le n \le M-1 \\ 0, & \text{otherwise} \end{cases}$$

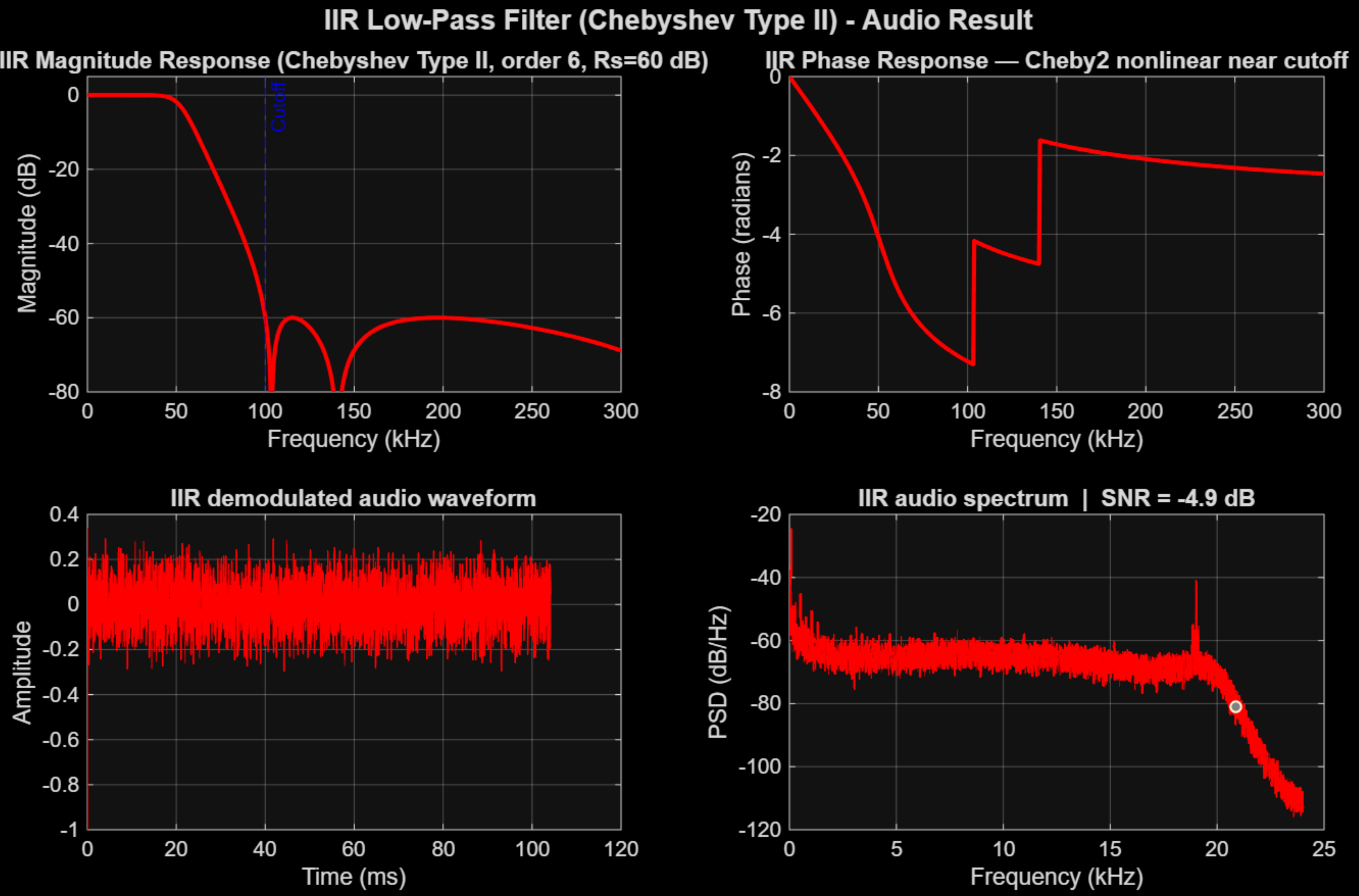
$$|H_a(j\Omega)|^2 = \frac{1}{1 + \varepsilon^2 \left(\dfrac{T_N(\frac{\Omega_s}{\Omega_p})}{T_N(\frac{\Omega_s}{\Omega})} \right)^2}$$

constant

$\sim 1/T_N(1/\Omega)$


FIR Low-Pass Filter - Audio Result
FIR Magnitude Response (Hamming, order 101)
Magnitude (dB)
Cutoff
Frequency (kHz)
FIR Phase Response - perfectly linear
Phase (radians)
Frequency (kHz)
FIR demodulated audio waveform
Amplitude
Time (ms)
FIR audio spectrum | SNR = -4.2 dB
PSD (dB/Hz)
Frequency (kHz)

Audio:


IIR Low-Pass Filter (Chebyshev Type II) - Audio Result
IIR Magnitude Response (Chebyshev Type II, order 6, Rs=60 dB)
Cutoff
Magnitude (dB)
Frequency (kHz)
IIR Phase Response — Cheby2 nonlinear near cutoff
Phase (radians)
Frequency (kHz)
IIR demodulated audio waveform
Amplitude
Time (ms)
IIR audio spectrum | SNR = -4.9 dB
PSD (dB/Hz)
Frequency (kHz)

Audio:

FIR vs IIR

Metric	FIR (Hamming)	IIR (Chebyshev)
Phase Response	Perfectly Linear	Non-linear (Phase Distortion)
Group Delay	Constant (All frequencies delayed equally)	Frequency-dependent (Varies near cutoff)
Computational Cost	High (Requires more coefficients/taps)	Low (Efficient, fewer coefficients)
Stability	Always Stable	Potentially Unstable (Feedback loops)

Task 4:

Enhance the audio quality by removing the noise floor using a Wiener Filter.

- Estimate the noise power spectrum Pnn(f) during a silent interval.

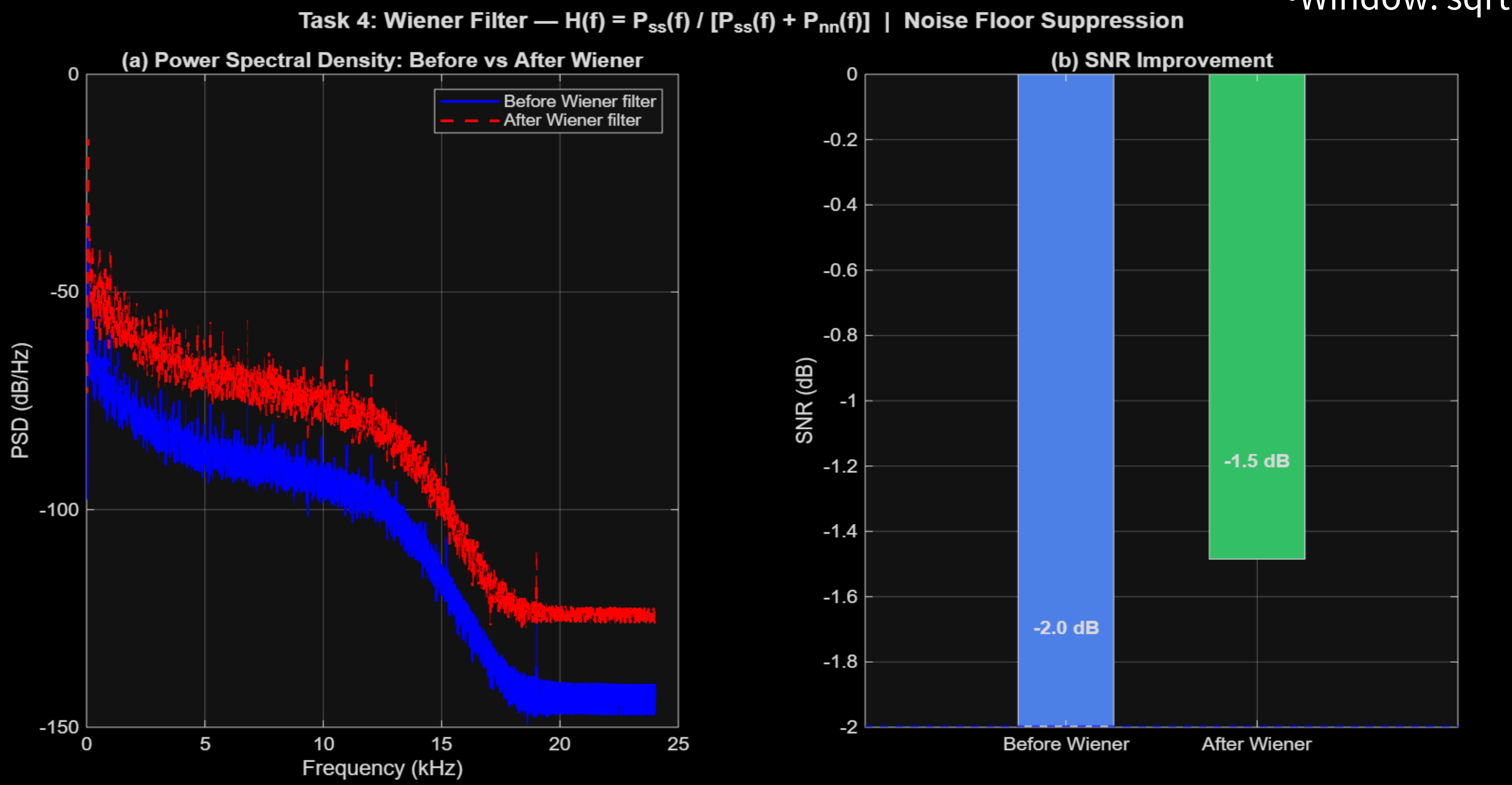
- Apply the Wiener filter transfer function: H(f) • Reconstruct the denoised audio signal.

$$H(f) = \frac{P_{ss}(f)}{P_{ss}(f) + P_{nn}(f)}$$

Before Wiener: noticeable background hiss
After Wiener: reduced noise and clearer audio
Trade-off: slight smoothing of signal

Frame Processing

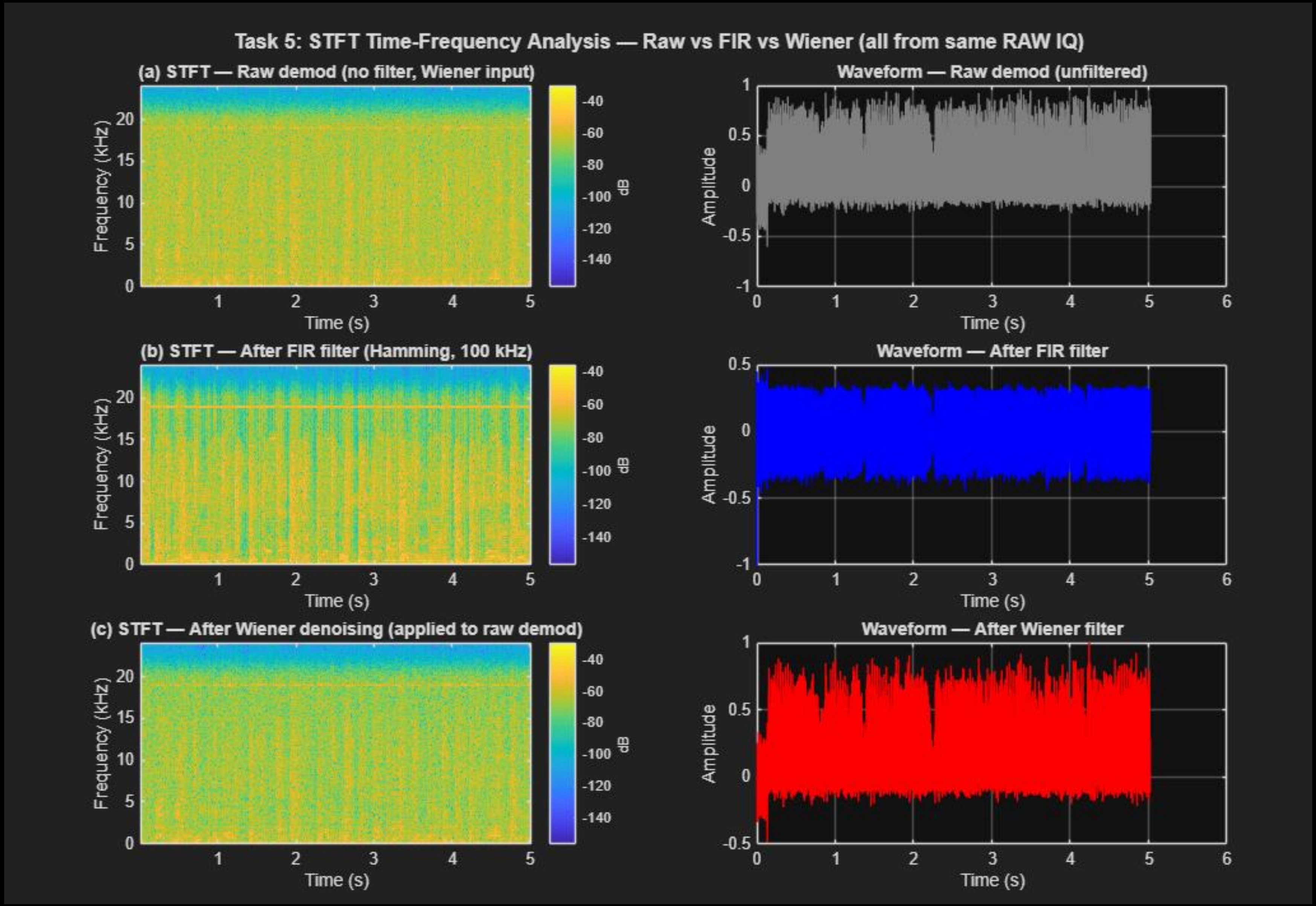
- Frame length: 2048 samples
- Hop size: 512 samples (75% overlap)
- Window: sqrt(Hamming window)



Task 5: Time-Frequency Analysis (STFT)

Perform Short-Time Fourier Transform (STFT) analysis to evaluate the time-varying frequency characteristics of the signal.

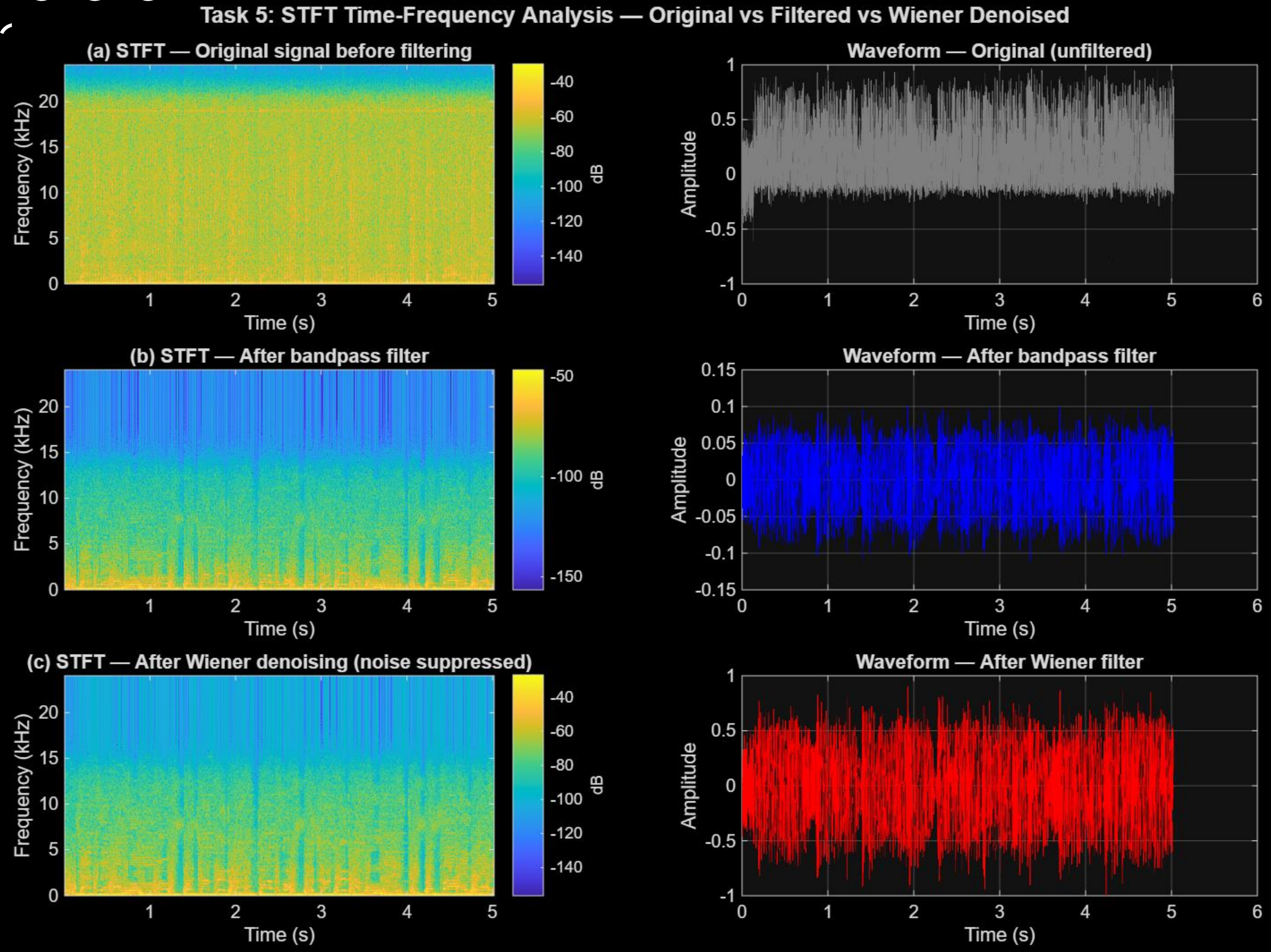
- Perform STFT on the signal before and after applying Wiener Filtering.
- Compare the spectrum and noise suppression observed by the filtering



Raw signal: wideband noise present
Filtered: reduced interference
Wiener: significant noise suppression

The line near 19 kHz corresponds to the FM stereo pilot tone.

Audio:



Here by bandpass meaning IIR filter was applied to see the difference between IIR and Wiener filter where cutoff is at 15kHz for the IIR Filter.

The cutoff was decided after looking into the Wiener filter and to observer side by side effects

Audio:

Code Pipeline

1.IQ Data Acquisition
The recorded SDR signal (IQ data) is loaded from a .mat file. Any invalid values (NaN/Inf) are removed and DC offset is eliminated to ensure a clean baseband signal.
2.FM Demodulation
The FM signal is demodulated using phase difference between consecutive samples:
demod = angle(iq(n) * conj(iq(n-1)))
This converts frequency variations into an amplitude signal (audio).
3.Downsampling
The signal is decimated in stages to reduce the sampling rate from 2.4 MHz to 48 kHz, making it suitable for audio processing.
4.FIR vs IIR Filtering
Two filters are applied separately:
•FIR (Hamming window): linear phase, better signal quality
•IIR (Chebyshev Type II): efficient but introduces phase distortion
Both outputs are compared using SNR and frequency response.

Code Pipeline

Bandpass filtering removes unwanted low/high frequency noise.
5.Wiener Filtering (Noise Reduction)
Noise is estimated from low-energy segments.
Wiener filter is applied using:
H(f) = Pss / (Pss + Pnn)
This reduces background noise while preserving useful signal.
6.Time-Frequency Analysis (STFT)
Spectrograms are used to analyze how signal energy varies over time and frequency for:
•Raw demodulated signal
•Filtered signal
•Wiener filtered output
8.Output
Final audio signals are saved as .wav files and visualized using waveform plots and spectrograms.

Conclusion

- FIR provides better performance for FM due to linear phase

- IIR is efficient but introduces phase distortion

- Wiener filter improves SNR by reducing noise adaptively

- SDR + DSP enables real-time signal enhancement

- FIR/IIR filters perform fixed frequency selection, whereas Wiener filtering is adaptive and based on signal and noise statistics

Audios

Raw Audio

Fir Filtered

IIR Filtered

Wiener Filtered

Thank You